

10 (a) State what is meant by contrast in an X-ray image.

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 [1]

(b) X-rays of intensity I_0 are incident normally on a structure, as shown in Fig. 10.1.

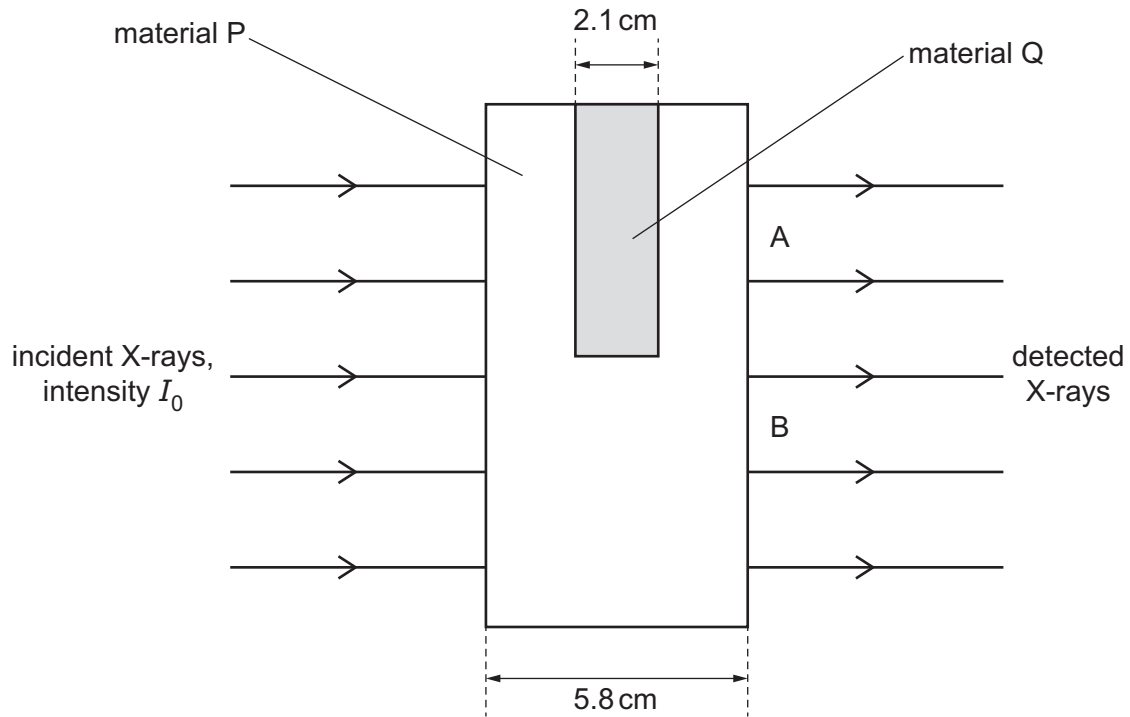


Fig. 10.1

Material P has a linear attenuation coefficient of 0.35 cm^{-1} .

The X-rays emerging from the structure in region A have an intensity of $0.053I_0$.

(i) Show that the intensity of the X-rays emerging in region B is $0.13I_0$.

[1]

